

EvidenceTree: Process-Rewarded Search over Retrieval Actions for Multimodal RAG

Anonymous submission

Abstract

Multimodal retrieval-augmented generation (RAG) hinges on a sequence of retrieval decisions — what to search, with which modality, and when to stop — yet current systems leave these decisions unsupervised: fixed pipelines apply one recipe to every question, outcome-only reinforcement learning misassigns credit across steps, and imitation-learned agents cannot recover once a retrieval goes wrong. Process reward models (PRMs) supply exactly the missing step-level signal, but existing PRMs score reasoning steps under a *fixed* context, an assumption that fails structurally in RAG where every retrieval action *modifies* the context. We present EvidenceTree, which closes this gap with three coupled components. (1) ET-PRM, a grounded, action-typed PRM for *state-modifying retrieval actions*, supervised by a dual-source step reward: unified-CLIP grounding of what each action actually retrieved, fused with a tree-level Monte Carlo outcome credit, and hardened by an answer-support gate that denies perfect labels to lucky, evidence-free answers. The PRM is trained generatively (rationale + score) and queried score-only at inference. (2) A Monte Carlo Tree Search that plans directly over the typed action space $\{\text{TEXT_SEARCH}, \text{IMAGE_SEARCH}, \text{ANSWER}\}$ with the PRM as the sole value signal, natively backtracking out of poisoned retrieval branches. (3) ETBench-Open, to our knowledge the first step-level annotated dataset of multimodal retrieval trajectories, built reproducibly on benchmark-provided corpora (InfoSeek, ScienceQA, GQA) under a leakage-proof rationale protocol. We specify the full evaluation design against published baselines; our own results are withheld pending training completion.

1 Introduction

Knowledge-intensive multimodal questions — “When did *this* building officially open?”, asked of a photograph — cannot be answered from model parameters alone: the entity must be recognized and its attributes retrieved from external corpora [8, 16]. Retrieval-augmented generation (RAG) is the standard approach to such questions [7, 23, 39], and its frontier has been moving from fixed retrieve-then-generate pipelines toward *agentic* systems that decide when and how to search [10, 37].

However, examining how these systems make their re-

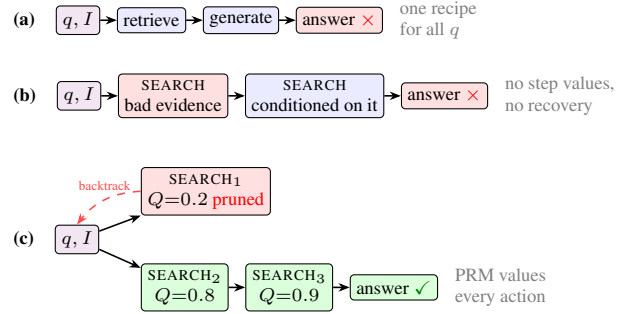


Figure 1: Three paradigms for multimodal RAG. (a) Fixed pipelines make no retrieval decisions. (b) Linear search agents decide but cannot *evaluate* or *undo* a decision: evidence from a bad retrieval conditions everything downstream (88.6% of wrong retrievals end in wrong answers for the linear agent of Chen et al. 10). (c) EvidenceTree scores every candidate retrieval action with a grounded PRM and searches over the action tree, pruning and backtracking out of bad branches.

trieval decisions reveals that the decisions themselves are essentially unsupervised. We identify three concrete challenges (Figure 1):

Challenge 1 — no evaluation of individual retrieval actions. Outcome-only reinforcement learning [34, 37] rewards a whole trajectory by its final answer: a useless search inside a lucky trajectory is reinforced, and a reasonable search inside a failed one is punished; the step-level signal needed to tell them apart does not exist in these systems. Imitation-learned agents [10] clone corrected search paths and never learn what makes an individual action good. Tellingly, in the analysis of Chen et al. [10], 29.6% of samples retrieve the right evidence yet still answer wrongly — retrieval correctness and answer correctness are far from aligned, and nothing in these systems can tell one step’s contribution from another’s.

Challenge 2 — no recovery from failed retrieval. Retrieval actions are *state-modifying*: retrieved evidence is appended to the context and conditions every subsequent decision. In a linear agent a single bad retrieval therefore poisons the remainder of the trajectory — 88.6% of the wrong-retrieval cases of Chen et al. [10] end in wrong answers — yet none of the existing multimodal search agents can back out of a branch once taken.

Challenge 3 — cross-modal planning is implicit. Whether to search with the *image* (to disambiguate an unnamed entity)

or with *text* (to fetch an attribute of a known entity) is a planning decision that current systems leave to prompt heuristics.

A natural remedy for Challenge 1 is a process reward model (PRM), which scores intermediate steps rather than outcomes [22, 33]. But transplanting existing PRMs fails for a structural reason: multimodal PRMs [35] and grounded PRMs [44] evaluate *reasoning steps under a fixed state* — the image and context are given once and never change — whereas in RAG the very object to be evaluated is an action that *changes* the state, whose quality depends on what it actually *returned*, and whose alternatives live on other branches of a search tree. No existing work simultaneously provides (i) grounded evaluation of retrieval actions, (ii) backtracking search over them, and (iii) cross-modal retrieval planning.

EvidenceTree is designed around this gap: one component per challenge, plus the dataset that makes the first component trainable:

- **ET-PRM: a grounded, action-typed PRM for retrieval actions** (Challenge 1). It scores a candidate action conditioned on the question, image, and accumulated evidence. Training labels fuse a graded *local grounding* signal — relevance of the action’s *retrieved results* to the question, measured in one unified CLIP space so text- and image-side evidence are comparable — with a *tree-level outcome credit*, the Monte Carlo success rate of all rollouts through the node. An *answer-support gate* denies perfect labels to answers that are correct but unsupported by any evidence (parametric lucky guesses). The PRM is trained *generatively* (rationale + score) for dense supervision and queried *score-only* at inference.
- **MCTS over the retrieval action space** (Challenge 2). Nodes are partial contexts $(q, I, \mathcal{E})$; edges are typed retrieval actions; selection is standard UCB1 with the PRM’s value as the sole quality signal. Because the tree branches over *actions*, abandoning a poisoned branch is a first-class operation rather than an impossibility.
- **A unified cross-modal action space** (Challenge 3). Both search actions query one CLIP-indexed corpus; the disambiguate-then-look-up chain (IMAGE_SEARCH names the entity, TEXT_SEARCH fetches its attribute) is expressible as two edges of the same tree, and preferring it is learned by the PRM rather than hard-coded.
- **ETBench-Open**. Training such a PRM requires supervision that does not exist: VisualPRM400K [35] contains almost no retrieval actions, RAG-ProGuide [43] is text-only, and DeepMMSearchVQA [28] has no step labels. We therefore construct ETBench-Open, a step-level annotated dataset of multimodal retrieval trajectories (target 50–80K trajectories with 150–300K step labels, depending on realized trajectory depth) built entirely on benchmark-provided corpora — no live web APIs, hence reproducible end to end — under a five-rule rationale quality-control protocol that excludes ground-truth leakage by construction.

Section 2 positions EvidenceTree against prior systems;

Sections 3 and 4 present the method and the dataset; Section 5 lays out the complete experimental design — benchmarks, baselines with their published numbers, and a pre-registered ablation grid. Our own numbers are deliberately withheld (marked “–”) pending completion of PRM training; every baseline number we print is quoted from the original publication cited in the caption.

2 Related Work

Tree search for multimodal RAG. RCTS [40] enriches its knowledge base with self-consistent reasoning contexts and performs MCTS with heuristic answer rewards to re-rank *retrieved examples* as in-context demonstrations: the search orders already-retrieved context rather than deciding what to retrieve. AR-MCTS [13] searches in *reasoning space*, using retrieval only to diversify expansion, and trains a PRM over reasoning steps. EvidenceTree differs from both in *where the tree lives*: our nodes are partial retrieval contexts and our edges are retrieval actions, so the search governs the retrieval process itself rather than its aftermath. In text-only RAG, ReasonRAG [43] builds process-supervised preference data via MCTS over an agentic action space, and TreePS-RAG [42] estimates step utility from descendant outcomes; neither is multimodal, neither trains a grounded PRM, and neither searches at inference time with a learned action-value model.

Process reward models. PRMs originated in mathematical reasoning [22, 33]. VisualPRM [35] extends process supervision to multimodal reasoning but scores steps under a *fixed* image context. GroundedPRM [44] verifies each reasoning step with external tools and fuses tool-based local signals with MCTS-derived outcome signals into rationale-enhanced generative supervision — for mathematical reasoning. We adopt its dual-source philosophy but transplant it to a setting it explicitly does not cover: *state-modifying* retrieval actions, where the object being verified is what an action *returned* rather than a self-contained reasoning statement, where grounding must span modalities in a single comparable space, and where credit must be assigned over a tree of alternative retrievals. Step-level rewards for *search* agents have recently appeared in text-only settings [2, 20, 38]; generative PRMs for web agents [3] score browser actions. None combine multimodal grounding, tree-level credit, and a learned generative PRM for retrieval actions.

Agentic multimodal search. MMSearch-R1 [37] trains an LMM with outcome-only GRPO to invoke image and text search on demand; DeepMMSearch-R1 [28] extends this with query-image cropping and a dedicated VQA dataset; VRAG-RL [34] adds visual perception actions. Closest to our setting, Learning-to-Search (DBAgent) [10] reformulates KB-VQA as a four-action decision process (ANSWER/IMAGE_SEARCH/TEXT_SEARCH/CAPTION)

(a) Inference: PRM-guided MCTS over retrieval actions

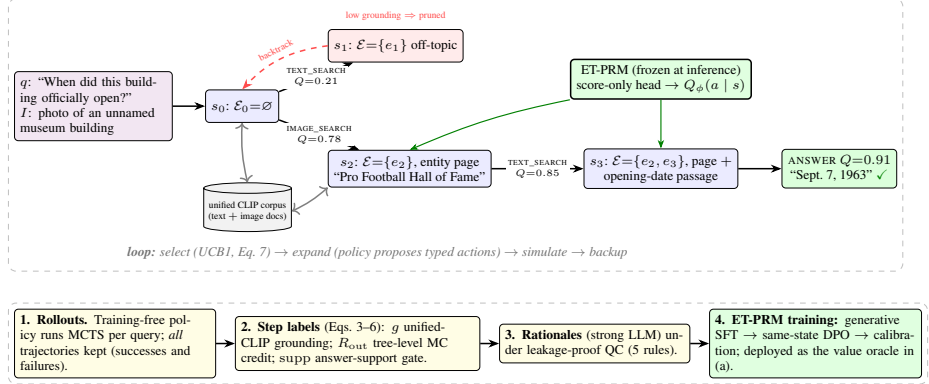


Figure 2: EvidenceTree overview. (a) The search tree lives in the *retrieval action space*: every edge is a typed action executed against the benchmark-provided corpus (one unified CLIP index for text and image documents), and every node is the resulting evidence state. The example illustrates the intended cross-modal plan on an entity question (the Q values are illustrative, not measured): a guessed TEXT_SEARCH retrieves off-topic evidence and is pruned by its low PRM value ($Q=0.21$); IMAGE_SEARCH disambiguates the entity ($Q=0.78$); a follow-up TEXT_SEARCH on the evidence-established name fetches the asked attribute ($Q=0.85$); the ANSWER edge closes the trajectory. UCB1 selection (Eq. 7) uses the PRM value as the only quality signal, so low- Q branches are abandoned — backtracking is a first-class operation. (b) The PRM is trained on ETBench-Open: MCTS rollouts are labeled per step with dual-source rewards and an answer-support gate, paired with QC-filtered rationales, and distilled into ET-PRM by generative SFT, same-state preference optimization, and calibration.

over a curated Wikipedia corpus and reaches state-of-the-art InfoSeek accuracy via supervised fine-tuning on rule-constructed trajectories. All of these are *linear* policies learned by imitation or outcome-only RL: they neither estimate step values nor backtrack. EvidenceTree is complementary — it learns to *evaluate* retrieval actions and to *plan* over them, keeping the policy training-free; DBAgent’s own error analysis (Section 1) documents precisely the failure modes that step-level evaluation with backtracking targets.

Step-level datasets. VisualPRM400K [35] contains almost exclusively mathematical reasoning steps; RAG-ProGuide [43] provides 13K process preference pairs for text-only RAG; DeepMMSearchVQA [28] provides multimodal search trajectories without step labels. ETBench-Open is, to our knowledge, the first step-level annotated dataset of *multi-modal retrieval actions*; at target scale it is designed to provide an order of magnitude more step-level labels than RAG-ProGuide’s 13K preference pairs.

3 Method

3.1 Problem Formulation

Given a question q and an optional image I , the system interacts with a fixed, benchmark-provided corpus $\mathcal{C}$ over multiple steps. The state after t steps is the partial retrieval context

$$s_t \triangleq (q, I, \mathcal{E}_t), \quad \mathcal{E}_t = \{e_1, \dots, e_{m_t}\}, \quad (1)$$

where $\mathcal{E}_t$ is the *evidence bundle* accumulated so far. At each step the system selects an action $a_t \in \mathcal{A}$; retrieval actions modify the state by appending their results, $s_{t+1} =$

$(q, I, \mathcal{E}_t \cup \text{obs}(a_t))$, and the terminal action emits an answer. A complete trajectory is $\tau = (s_0, a_0, s_1, \dots, a_T)$ with $a_T = \text{ANSWER}(y)$. The goal is to choose the sequence of retrieval actions that maximizes the probability that y matches the gold answer y^* , under a fixed rollout budget. Unlike reasoning-step search [13, 35], every non-terminal action here is *state-modifying* and *irreversible within its branch*: evidence, once retrieved, conditions all subsequent decisions on that path (Challenge 2). Tree search provides the escape hatch — a branch poisoned by a bad retrieval can be abandoned rather than conditioned on. Notation is summarized in Appendix G.

3.2 Retrieval Action Space

We type actions by the *modality of the retrieval query*, not of the result:

$$\mathcal{A} = \{\text{TEXT_SEARCH}(x), \text{IMAGE_SEARCH}(I), \text{ANSWER}(y)\}. \quad (2)$$

Both search actions query the *same* unified index: every corpus document — text chunk or image — is embedded with one CLIP model [30], so a text query or an image query retrieves whichever side of the corpus is most similar. This addresses Challenge 3 structurally rather than heuristically: for entity questions where the entity is not named in q (“this bird”, “this building”), IMAGE_SEARCH matches the query image against the corpus to *disambiguate the entity*, after which TEXT_SEARCH retrieves the asked attribute using the evidence-established entity name (Figure 2a) — and *whether* to prefer this chain on a given state is learned by the PRM, not hard-coded. We deliberately do not type actions by result modality (a 2×2 split dilutes the visit counts of each

branch under small rollout budgets, and the retriever already decides result modality by similarity), do not include compound parallel-search actions (tree expansion already provides multi-candidate exploration, and multi-source fusion is expressible as a serial trajectory), and do not include state-invariant actions such as captioning (an edge that does not change the observable state would receive no grounding signal and dilute tree-level credit; captioning is folded into the policy prompt instead). Appendix A details these choices.

3.3 Dual-Source Step Reward

Addressing Challenge 1 requires a per-step label that neither trusts the outcome blindly (credit misassignment) nor ignores it (retrieval relevance alone does not imply usefulness). Each step of a trajectory therefore receives a graded label in $[0, 1]$ fusing two complementary signals.

Local grounding. Grounding measures *how relevant the results retrieved by this step are to the question* — deliberately scoring the result side rather than the query side, since the query image of IMAGE_SEARCH is fixed by the state and carries no signal about whether the step helped. All results are scored in one CLIP space against the same anchor:

$$g(a_t, s_t) = \max_{r \in \text{obs}(a_t)} \phi_{m(r)} \left(\cos(E_T(q), E_{m(r)}(r)) \right), \quad (3)$$

where E_T and E_I are the CLIP text and image towers, $m(r)$ is the modality of result r , and ϕ_m is a per-modality-pair linear calibration band mapping the raw cosine onto a common $[0, 1]$ relevance scale (same-modality cosines run systematically higher than cross-modal ones). Empty results give $g = 0$. Using a *single* model for both sides is essential: two different encoders would put text- and image-side evidence on different cosine scales, and the PRM would mistake the scale gap for a quality gap, systematically biasing action selection. All grounding scores are graded, not binary — retrieval quality in open corpora is not a 0/1 property (cf. the binary verification of Zhang et al. 44).

Tree-level outcome credit. Outcome credit estimates each node’s true success potential. For a node reached by the action prefix $\pi_t = (a_0, \dots, a_t)$ within one question’s search tree, let $\mathcal{T}(\pi_t)$ denote all complete rollouts passing through it. Then

$$R_{\text{out}}(s_t, a_t) = \frac{1}{|\mathcal{T}(\pi_t)|} \sum_{\tau \in \mathcal{T}(\pi_t)} \mathbb{K}[y(\tau) \approx y^*], \quad (4)$$

a Monte Carlo estimate of the success rate *from this node onward*. This is crucially different from uniformly smearing a trajectory’s final reward over its steps — uniform credit rewards the bad steps of lucky trajectories and punishes the good steps of unlucky ones, whereas Eq. (4) lets two children

of the same node receive sharply different labels (cf. shortest-path estimation in Zhang et al. 43; descendant-outcome utilities in Zhang et al. 42). Answer matching $\mathbb{K}[\cdot \approx y^*]$ follows each benchmark’s official protocol (relaxed numeric-range matching for InfoSeek VALUE questions; Appendix C).

Fusion and the answer-support gate. Non-terminal steps fuse the two sources linearly:

$$R(s_t, a_t) = \alpha g(a_t, s_t) + (1 - \alpha) R_{\text{out}}(s_t, a_t), \quad (5)$$

with $\alpha = 0.5$ by default (sensitivity in Section 5.3). Terminal ANSWER steps have no retrieval result to ground, and labeling them by outcome alone creates a subtle poison that we observed directly in our construction pilot (Appendix F, Case 2): a *parametric lucky guess* — an answer that happens to be correct although nothing in the evidence supports it — would receive a perfect label, teaching the PRM that evidence-free answering is optimal. We therefore gate the outcome by an *answer-support* verdict $\text{supp}(y, \mathcal{E}_T) \in [0, 1]$, judged by an LLM over the accumulated evidence (four-way rubric: SUPPORTED/PARTIAL/UNSUPPORTED $\mapsto 1/0.5/0$, plus NOT_REQUIRED $\mapsto$ no gating for questions answerable from the question and image alone by perception or logic — entailment-gating those would systematically punish correct reasoning answers; Appendix B and Case 3):

$$R_{\text{ans}}(s_T) = R_{\text{out}}(s_T, a_T) \cdot (\gamma + (1 - \gamma) \text{supp}(y, \mathcal{E}_T)), \quad (6)$$

with floor $\gamma = 0.3$. The multiplicative form keeps wrong answers at zero regardless of spurious support; the floor preserves the ordering *wrong* < *correct-but-unsupported* ($\approx \gamma$) < *correct-and-supported* (≈ 1) and buffers verifier noise. Correct-but-unsupported answers are additionally flagged — not discarded — as ready-made rejected sides for preference training (Section 3.4).

3.4 ET-PRM: Architecture and Training

Architecture. ET-PRM is initialized from VisualPRM-8B [35] (itself built on InternVL2.5-8B [9]), inheriting multimodal inductive biases, with an action-conditional value readout. The input serializes the state and the candidate action: question and image, the evidence bundle, and the candidate action’s type and parameters; the output is a rationale followed by a score token.

Generative training, score-only inference. Following the generative reward modeling paradigm [1, 41], ET-PRM is trained to emit a natural-language rationale *and* a score. This is a deliberate choice with three payoffs at our data scale: each sample supervises dozens of tokens rather than one scalar (sample efficiency); the model must articulate *why* an action deserves its score, exposing spurious shortcuts (anti-reward-hacking); and failed training runs can be diagnosed by reading rationales. At inference the rationale is skipped entirely — only the score-token logit is read — so MCTS pays no generation latency.

Three training stages. (1) *SFT*: next-token prediction on ETBench-Open (state, action, rationale, score). (2) *Action-typed preference optimization*: DPO [31] on same-state contrastive pairs mined from the search trees — (successful sibling action, failed action), (TEXT_SEARCH, IMAGE_SEARCH) under the same state, and (supported answer, unsupported-correct answer), the latter using the flagged samples of Eq. (6). Pair construction exploits the same node keys as Eq. (4), so the contrast dimension is “same state, different action” — matching the decision the searcher actually faces — rather than the step-position contrasts used for reasoning PRMs. (3) *Calibration*: temperature/Platt scaling of the score logit on held-out data, so that Q values entering UCB1 have probabilistic meaning.

3.5 PRM-Guided MCTS over Retrieval Actions

Search follows the four canonical MCTS phases with the PRM as the value oracle. *Selection* descends by UCB1 [4, 19]:

$$a^\dagger = \arg \max_a Q(a | s) + c \sqrt{\frac{\ln N(s)}{N(s, a)}}, \quad (7)$$

where Q is the running mean of backed-up rewards, seeded for unvisited edges by the PRM prior $Q_\phi(a | s)$, and $c = 1.0$ by default. We use *no* additional novelty or modality-coverage bonus: the PRM’s Q is the sole quality signal, and hard-coded exploration preferences would both dilute it and add tuning burden (Appendix A). *Expansion* asks a training-free policy VLM to propose candidate actions (with anti-fabrication and retrieval-chain constraints in the prompt; Appendix E), executes them against the local index, and keeps the top- k by the PRM prior Q_ϕ . *Simulation* rolls out greedily to a forced answer at depth $d_{\max}$. *Backup* propagates the trajectory reward to the root. Search stops early when the best root path exceeds a quality threshold, or when the roll-out budget P is exhausted (defaults: $P=10$, $d_{\max}=3$, $k=3$). Algorithm 1 summarizes the loop.

4 The ETBench-Open Dataset

ETBench-Open is built by a four-step pipeline (Figure 2b), entirely on corpora that ship with the benchmarks: a Wikipedia entity corpus for InfoSeek (chunked full pages of the 1,794 validation entities), the lecture/context corpus for ScienceQA, and scene-graph-derived textual documents paired with images for GQA. This design has three consequences we consider essential for a paper about *decision quality*: reproducibility (no live-web drift), causal cleanliness (performance differences cannot be explained by “the web changed”), and auditability (every evidence item in every label is a corpus document with a stable identifier). *Step 1 (rollout)* runs the training-free policy through MCTS with

Algorithm 1 EvidenceTree inference

Require: question q , image I , corpus index, budget P , depth $d_{\max}$

- 1: root $\leftarrow (q, I, \emptyset)$
- 2: **for** $t = 1 \dots P$ **do**
- 3: $s \leftarrow \text{SELECT}(\text{root})$ by Eq. (7)
- 4: $\mathcal{A}_c \leftarrow$ policy proposals at s ; execute retrievals
- 5: keep top- k children of s by PRM prior $Q_\phi(a | s)$
- 6: $\tau \leftarrow \text{SIMULATE}$ greedily to ANSWER or $d_{\max}$
- 7: BACKUP(τ , $R_{\text{PRM}}(\tau)$)
- 8: **if** $\max_\tau R_{\text{PRM}}(\tau) > \eta_{\text{stop}}$ **then break**
- 9: **end if**
- 10: **end for**
- 11: **return** answer of the highest-value root path

early stopping disabled, so the tree branches widely; all deduplicated trajectories are kept — failures included, since low-scored steps are the PRM’s negative curriculum. *Step 2 (labels)* computes Eqs. (3)–(6) per step. *Step 3 (rationales)* has a strong LLM explain each score. *Step 4 (QC)* applies the only three drop rules — trajectory length outside $[2, 8]$, grounding–outcome self-contradiction ($|g - R_{\text{out}}| > 0.7$), and rationale QC failure — and splits train/validation by query so no question leaks across splits.

Rationale quality control. Generative PRM training is only as good as its rationales. Each rationale must (a) cite at least one evidence identifier visible in the sample, (b) name the action type, (c) be 30–150 tokens, (d) contain *no ground-truth leakage* — no evaluation phrasing, and no gold-answer string that is absent from the sample’s visible evidence, question, and action input (quoting the answer *from cited evidence* is legitimate; knowing it without evidence is leakage — rationales must argue *ex-ante*, as at inference the PRM has no oracle), and (e) end with a VERDICT tag that must not contradict the numeric score (preventing say-one-thing-score-another training pairs). Rules (d)/(e) adapt the trajectory-realism constraints of Chen et al. [10] from policy imitation to reward-model supervision. Full prompts are given in Appendix E.

Scale and status. Target scale is 50–80K trajectories (150–300K step labels, depending on the trajectory depths realized after the retrieval-chain fixes of Appendix D) across the three benchmarks. Construction statistics of a completed 100-query-per-benchmark pilot — including the step-label and support-verdict distributions that motivated the support gate — are reported in Appendix D, and worked success/failure cases from the pilot in Appendix F. Full-scale construction and PRM training are in progress; accordingly, all Evidence-Tree result cells in Section 5 are marked “–”.

5 Experimental Design

5.1 Setup

Benchmarks and metrics. We evaluate on three benchmarks with complementary demands. **InfoSeek** [8]: fine-grained entity knowledge VQA over a Wikipedia corpus; validation split, official relaxed accuracy (string questions: relaxed exact match over the gold set; numeric questions: any predicted number inside the gold range), reported on Unseen-Question / Unseen-Entity / All. **ScienceQA** [27]: multiple-choice science questions; test split accuracy. **GQA** [17]: compositional VQA over real images; test-dev balanced accuracy. ScienceQA and GQA are not knowledge-seeking in the InfoSeek sense; we include them to probe two specific behaviors — whether the PRM learns *when not to retrieve* (ScienceQA reasoning questions) and whether image-query retrieval provides clean perception evidence (GQA scene-graph corpus) — and we analyze them separately from the open-domain headline results.

Implementation. Policy: Qwen2.5-VL-7B-Instruct [6], training-free (prompt in Appendix E). Retrieval: BM25 [32] and dense CLIP indices (FAISS [18]) built offline per benchmark corpus; text top-3, image top-1 following Chen et al. [10]. PRM: VisualPRM-8B fine-tuned with LoRA [15] through the three stages of Section 3.4. Search: $P=10$, $d_{\max}=3$, $k=3$, $c=1.0$, $\alpha=0.5$, $\gamma=0.3$. All results (once reported) will be means over 3 seeds.

5.2 Baselines

We compare against four families: (i) zero-shot MLLMs (no retrieval); (ii) fixed-pipeline multimodal RAG; (iii) reflection- or process-enhanced RAG; (iv) agentic / decision-based search including the strongest published InfoSeek system, DBAgent [10]; plus PRM-with-Best-of- N controls (frozen VisualPRM-8B reranking N sampled trajectories) that isolate the contribution of tree search over sample-and-rerank. Baseline numbers in Tables 1–3 are quoted from the cited publications; rows we will run ourselves under our corpus protocol are marked accordingly.

5.3 Ablation Design

Table 4 pre-registers the ablation grid, one block per design decision. **A1** tests whether both reward sources are necessary ($\alpha \in \{0, 0.2, 0.5, 0.8, 1\}$) and whether the support gate matters (Eq. 6 on/off). **A2** varies the exploration constant ($c \in \{0, 0.5, 1, 2\}$; $c=0$ is greedy PRM). **A3** accumulates the PRM training stages (SFT / +DPO / +calibration) against the frozen VisualPRM-8B zero-shot scorer. **A4** grows the action space (TEXT_SEARCH-only $\rightarrow$ +IMAGE_SEARCH). **A5** varies the search budget ($d_{\max} \in \{2, 3, 5\}$, $P \in \{4, 10, 20\}$) against Best-of- N at matched PRM-call budgets — the decisive test of whether *tree* search, rather than PRM rerank-

Method	Backbone	Un-Q	Un-E	All
<i>Zero-shot MLLMs (no retrieval)</i>				
BLIP-2 [21]	Flan-T5-XXL	12.7	12.3	12.5
InstructBLIP [12]	Flan-T5-XXL	8.9	7.4	8.1
LLaVA-1.5 [25]	Vicuna-7B	9.6	9.4	9.5
GPT-4V [§]	—	15.0	14.3	14.6
Qwen2.5-VL [§] [6]	7B	22.8	24.1	23.7
<i>Retrieval-augmented (fixed pipeline / reflective)</i>				
Wiki-LLaVA [7]	LLaVA-1.5-7B	30.1	27.8	28.9
PreFLMR [23]	RA-VQAv2	—	—	30.7
EchoSight [39]	LLaMA-3-8B	—	—	31.3
RoRA-VLM [29]	Vicuna-7B	25.1	27.3	—
ReflectiVA [11]	LLaMA-3.1-8B	40.4	39.8	40.1
Wiki-PRF [14]	Qwen2.5-VL-7B	43.3	42.7	42.8
<i>Agentic / decision-based search</i>				
Wiki-R1 [36]	Qwen2.5-VL-7B	47.8	—	44.1
DBAgent [10]	Qwen2.5-VL-7B	46.5	51.0	49.9
VisualPRM-8B + BoN [‡]	Qwen2.5-VL-7B	—	—	—
EvidenceTree (ours)	Qwen2.5-VL-7B	—	—	—

Table 1: InfoSeek (validation, official relaxed accuracy). Baseline numbers are quoted from the cited papers (BLIP-2/InstructBLIP from the official evaluation kit of Chen et al. 8; Wiki-LLaVA and LLaVA-1.5 from Caffagni et al. 7, Table 2; EchoSight from Yan and Xie 39; PreFLMR from Lin et al. 23, Table 7, whose authors note a pre-release InfoSeek setup; RoRA-VLM from Qi et al. 29; ReflectiVA from Cocchi et al. 11; Wiki-PRF from Hong et al. 14; Wiki-R1 from Wiki-R1 authors 36, unseen-entity split not reported; DBAgent (SFT on InfoSeek trajectories) from Chen et al. 10, Table 1. §: zero-shot numbers as compiled in Chen et al. 10, Table 1.) ‡: control we run under the identical corpus protocol. “—”: training in progress.

ing alone, carries any gains. **A6** evaluates robustness under corpus poisoning [26]: 5–20 adversarial image–text pairs injected per query; we measure the answer flip rate with and without PRM-guided backtracking.

5.4 Analysis Protocol

Beyond headline accuracy we will report: (i) the retrieval-correctness $\times$ answer-correctness contingency table, whose off-diagonal mass motivates step-level evaluation (cf. 29.6%/88.6% in Chen et al. 10); (ii) PRM score–outcome correlation (Spearman/AUC) against the frozen VisualPRM-8B diagnostic baseline; (iii) per-action-type PRM calibration curves; and (iv) corpus-scale robustness (10K $\rightarrow$ 100K documents).

6 Conclusion

EvidenceTree moves grounded process supervision from fixed-state reasoning to state-modifying retrieval actions: a PRM that evaluates what a retrieval *returned* (in one unified multimodal embedding space), credits each node by the

Method	Backbone	Acc.
Human [27]	–	88.40
MM-CoT [45]	UnifiedQA-738M	91.68
LLaVA (fine-tuned) [24]	Vicuna-13B	90.92
Zero-shot [†]	Qwen2-VL-7B	80.33
Zero-shot [†]	InternVL2-8B	93.00
Vanilla RAG [†]	Qwen2-VL-7B	86.68
RCTS [40]	Qwen2-VL-7B	91.44
RCTS [40]	InternVL2-8B	94.20
VisualPRM-8B + BoN [‡]	Qwen2.5-VL-7B	–
EvidenceTree (ours)	Qwen2.5-VL-7B	–

Table 2: ScienceQA (test accuracy). Baseline numbers quoted from the cited papers (MM-CoT from Zhang et al. 45; LLaVA from Liu et al. 24, Table 7; †: as measured in the RCTS paper Yang et al. 40 under its evaluation protocol; RCTS from the same table. AR-MCTS Dong et al. 13 does not evaluate on ScienceQA; methods reporting only the IMG subset, e.g. LLaVA-1.5 at 66.8, are omitted for protocol consistency.)

Method	Backbone	Acc.
BLIP-2* [21]	Vicuna-13B	41.0
InstructBLIP [12]	Vicuna-7B	49.2
Qwen-VL-Chat [5]	Qwen-7B	57.5
Qwen-VL [5]	Qwen-7B	59.3
LLaVA-1.5 [25]	Vicuna-7B	62.0
LLaVA-1.5 [25]	Vicuna-13B	63.3
EvidenceTree (ours)	Qwen2.5-VL-7B	–

Table 3: GQA (test-dev balanced accuracy). Baseline numbers quoted from the cited papers (Qwen-VL rows from Bai et al. 5; LLaVA-1.5 and InstructBLIP from Liu et al. 25 and Dai et al. 12; *: zero-shot number as reported in Liu et al. 25, Table 1. Existing multimodal RAG and tree-search systems do not report GQA; we include it to probe perception-evidence retrieval over a scene-graph corpus, not as an open-domain RAG benchmark.)

Monte Carlo success of the subtree behind it, refuses to reward evidence-free lucky answers, and guides a standard UCB1 tree search that can back out of poisoned branches. The accompanying ETBench-Open provides the first step-level supervision for multimodal retrieval actions, with a leakage-proof rationale protocol. The experimental grid is fully specified; results will populate the “–” cells upon training completion.

References

- [1] Zachary Ankner, Mansheej Paul, Brandon Cui, Jonathan D. Chang, and Prithviraj Ammanabrolu. Critique-out-loud reward models. *arXiv preprint arXiv:2408.11791*, 2024.
- [2] Anonymous. PPR: Principle process reward for search agents. OpenReview preprint, 2026.

Ablation	Variant	InfoSeek	SQA	GQA
A1 reward	local only ($\alpha=1$)	–	–	–
	outcome only ($\alpha=0$)	–	–	–
	no support gate	–	–	–
A2 search	greedy ($c=0$)	–	–	–
	$c \in \{0.5, 2.0\}$	–	–	–
A3 PRM	frozen VisualPRM	–	–	–
	SFT only	–	–	–
	SFT+DPO	–	–	–
A4 actions	text-only	–	–	–
	+ image search	–	–	–
A5 budget	BoN, matched calls	–	–	–
	$P=4/20$	–	–	–
	$d_{\max}=2/5$	–	–	–

Table 4: Pre-registered ablation grid (accuracy; “–” = training in progress). A6 (poisoning robustness) reports answer flip rate and is tabulated separately.

- [3] Anonymous. WebArbiter: A generative process reward model for web agents. *arXiv preprint arXiv:2601.21872*, 2026.
- [4] Peter Auer, Nicolò Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47:235–256, 2002.
- [5] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- [6] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- [7] Davide Caffagni, Federico Cocchi, Nicholas Moratelli, Sara Sarto, Marcella Cornia, Lorenzo Baraldi, and Rita Cucchiara. Wiki-LLaVA: Hierarchical retrieval-augmented generation for multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 1818–1826, 2024.
- [8] Yang Chen, Hexiang Hu, Yi Luan, Haitian Sun, So-ravit Changpinyo, Alan Ritter, and Ming-Wei Chang. Can pre-trained vision and language models answer visual information-seeking questions? In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 14948–14968, 2023.

- [9] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2024. InternVL2.5.
- [10] Zhuohong Chen, Zhenxian Wu, Yunyao Yu, Hangrui Xu, Zirui Liao, Zhifang Liu, Xiangwen Deng, Pen Jiao, and Haoqian Wang. Learning to search: A decision-based agent for knowledge-based visual question answering. *arXiv preprint arXiv:2604.07146*, 2026.
- [11] Federico Cocchi, Nicholas Moratelli, Marcella Cornia, Lorenzo Baraldi, and Rita Cucchiara. Augmenting multimodal LLMs with self-reflective tokens for knowledge-based visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [12] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instruct-BLIP: Towards general-purpose vision-language models with instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [13] Guanting Dong, Chenghao Zhang, Mengjie Deng, Yutao Zhu, Zhicheng Dou, and Ji-Rong Wen. Progressive multimodal reasoning via active retrieval. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. AR-MCTS, arXiv:2412.14835.
- [14] Yuyang Hong, Jiaqi Gu, Qi Yang, Lubin Fan, Yue Wu, Ying Wang, Kun Ding, Shiming Xiang, and Jieping Ye. Knowledge-based visual question answer with multimodal processing, retrieval and filtering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. Wiki-PRF, arXiv:2510.14605.
- [15] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [16] Hexiang Hu, Yi Luan, Yang Chen, Urvashi Khandelwal, Mandar Joshi, Kenton Lee, Kristina Toutanova, and Ming-Wei Chang. Open-domain visual entity recognition: Towards recognizing millions of wikipedia entities. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [17] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [18] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.
- [19] Levente Kocsis and Csaba Szepesvári. Bandit based Monte-Carlo planning. In *Proceedings of the 17th European Conference on Machine Learning (ECML)*, 2006.
- [20] Yongqi Leng et al. DecEx-RAG: Boosting agentic retrieval-augmented generation with decision and execution optimization via process supervision. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2025.
- [21] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [22] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- [23] Weizhe Lin, Jingbiao Mei, Jinghong Chen, and Bill Byrne. PreFLMR: Scaling up fine-grained late-interaction multi-modal retrievers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5294–5316, 2024.
- [24] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [25] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [26] Yinuo Liu, Zenghui Yuan, Guiyao Tie, Jiawen Shi, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong. Poisoned-MRAG: Knowledge poisoning attacks to multimodal retrieval augmented generation. *arXiv preprint arXiv:2503.06254*, 2025.
- [27] Pan Lu, Swaroop Mishra, Tony Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. Learn to explain: Multimodal reasoning via thought chains for science question answering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [28] Kartik Narayan, Yang Xu, Tian Cao, Kavya Nerella, Vishal M. Patel, Navid Shiee, Peter Grasch, Chao Jia, Yinfei Yang, and Zhe Gan. DeepMMSearch-R1: Empowering multimodal LLMs in multimodal web search. *arXiv preprint arXiv:2510.12801*, 2025.

- [29] Jingyuan Qi, Zhiyang Xu, Rulin Shao, Zihao Lin, Yang Chen, Di Jin, Yu Cheng, Qifan Wang, and Lifu Huang. RoRA-VLM: Robust retrieval-augmented vision language models. *arXiv preprint arXiv:2410.08876*, 2024.
- [30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [31] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [32] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.
- [33] Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439, 2024.
- [34] Qiuchen Wang, Ruixue Ding, Yu Zeng, Zehui Chen, Lin Chen, Shihang Wang, Pengjun Xie, Fei Huang, and Feng Zhao. VRAG-RL: Empower vision-perception-based RAG for visually rich information understanding via iterative reasoning with reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [35] Weiyun Wang, Zhangwei Gao, Lianjie Chen, Zhe Chen, Jinguo Zhu, Xiangyu Zhao, Yangzhou Liu, Yue Cao, Shenglong Ye, Xizhou Zhu, Lewei Lu, Haodong Duan, Yu Qiao, Jifeng Dai, and Wenhai Wang. VisualPRM: An effective process reward model for multimodal reasoning. *arXiv preprint arXiv:2503.10291*, 2025.
- [36] Wiki-R1 authors. Wiki-R1: Incentivizing multimodal reasoning for knowledge-based VQA via data and sampling curriculum. In *International Conference on Learning Representations (ICLR)*, 2026. *arXiv:2603.05256*.
- [37] Jinming Wu, Zihao Deng, Wei Li, Yiding Liu, Bo You, Bo Li, Zejun Ma, and Ziwei Liu. MMSearch-R1: Incentivizing LMMs to search. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. *arXiv:2506.20670*.
- [38] Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, Xinya Du, and Zhiyu Chen. HiPRAG: Hierarchical process rewards for efficient agentic retrieval augmented generation. *arXiv preprint arXiv:2510.07794*, 2025.
- [39] Yibin Yan and Weidi Xie. EchoSight: Advancing visual-language models with wiki knowledge. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1538–1551, 2024.
- [40] Qi Yang, Chenghao Zhang, Lubin Fan, Kun Ding, Jieping Ye, and Shiming Xiang. Re-ranking reasoning context with tree search makes large vision-language models stronger. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. RCTS, *arXiv:2506.07785*.
- [41] Lunjun Zhang, Arian Hosseini, Hritik Bansal, Mehran Kazemi, Aviral Kumar, and Rishabh Agarwal. Generative verifiers: Reward modeling as next-token prediction. *arXiv preprint arXiv:2408.15240*, 2024.
- [42] Tianhua Zhang, Kun Li, Junan Li, Yunxiang Li, Hongyin Luo, Xixin Wu, James Glass, and Helen Meng. TreePS-RAG: Tree-based process supervision for reinforcement learning in agentic RAG. *arXiv preprint arXiv:2601.06922*, 2026.
- [43] Wenlin Zhang, Xiangyang Li, Kuicai Dong, Yichao Wang, Pengyue Jia, Xiaopeng Li, Yingyi Zhang, Derong Xu, Zhaocheng Du, Huifeng Guo, Ruiming Tang, and Xiangyu Zhao. Process vs. outcome reward: Which is better for agentic RAG reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [44] Yao Zhang, Yu Wu, Haowei Zhang, Weiguo Li, Haokun Chen, Jingpei Wu, Guohao Li, Zhen Han, and Volker Tresp. GroundedPRM: Tree-guided and fidelity-aware process reward modeling for step-level reasoning. *arXiv preprint arXiv:2510.14942*, 2025.
- [45] Zhuosheng Zhang, Aston Zhang, Mu Li, Hai Zhao, George Karypis, and Alex Smola. Multimodal chain-of-thought reasoning in language models. *Transactions on Machine Learning Research (TMLR)*, 2024.

A Action Space: Rejected Designs

Why not a 2×2 (query $\times$ result) action split. Under a rollout budget of $P=10$, four search actions halve per-branch visit counts; result modality is better decided by the retriever’s similarity in the unified index; and grounding already scores by the *actual* modality of each returned result (Eq. 3), so the action type need not carry it.

Why no PARALLEL_SEARCH. Tree expansion already explores multiple candidates (OR-semantics), and AND-semantics (fuse evidence from two sources) is expressible serially — search A , then B ; the bundle accumulates. A compound action would need bespoke grounding (means over sub-actions), conflict with tree-level credit, and dilute the budget.

Why no CAPTION action. Captioning does not modify the observable state: an edge with an unchanged state has no retrieval result to ground (g undefined), creates parent–child duplicates that dilute Eq. (4), and burns budget. The policy VLM sees the image directly, so caption-style verbalization is folded into the proposal prompt (Appendix E); cf. the caption action of Chen et al. [10], which their inference restricts to an optional caption emitted before the final action to help form a better text query.

Why no modality-coverage bonus in UCB1. With three actions there is little to “cover”; a λ -weighted novelty term adds three hyperparameters and competes with the PRM’s Q for the same decision, diluting the learned signal. Deferred visual operations (CROP/ZOOM) await a needs analysis of fine-grained visual queries.

B Answer-Support Judging

The judge receives the question, the accumulated evidence, and the proposed answer, and returns one of four verdicts. SUPPORTED: the evidence contains or entails the answer (for multiple choice, entailing the *content* of the chosen option suffices; an absence answer such as “no” is supported when same-image evidence describes the scene and lacks the item). PARTIAL: substantive part unconfirmed. UNSUPPORTED: the answer rests on external facts absent from the evidence. NOT_REQUIRED: derivable from the question and visible image alone by perception/logic/commonsense — never applicable to fine-grained entity facts (species data, dates, measurements). The fourth verdict maps to *no gating* (outcome-only), which repairs the systematic penalty that entailment-gating inflicts on reasoning-type multiple-choice answers (Appendix F, Case 3), while leaving the anti-guess gate fully active on knowledge-seeking questions. Raw verdicts ship with the dataset for auditability. An offline lexical fallback (answer content-word/number coverage in the evidence, with unmatched numbers vetoing) supports API-free smoke runs; unparseable or unverifiable cases return no verdict and fall back to outcome-only rather than fabricating a value. The judge prompt is given verbatim in Appendix E.

C Dataset Schema and Quality Control

Each released sample carries: identifiers (sample/trajectory/query); the state snapshot before the action (actions so far, evidence bundle); the typed action; the observed results; labels g , R_{out} (with the count of trajectories through the node), supp with its raw verdict, the fused score; the rationale, its verdict tag, QC status; and generation provenance (policy model and deployment, retriever, MCTS configuration, seeds, pipeline commit). Hard drop rules (the only ways a sample is excluded): trajectory length outside $[2, 8]$; non-answer steps with $|g - R_{\text{out}}| > 0.7$ (label self-contradiction); rationale QC failure after three regenerations. Failed trajectories are retained — their low-scored steps are the PRM’s negative curriculum — and correct-but-unsupported answers are retained with flags as DPO rejected candidates. Validation enforces score-fusion identities, per-field ranges, the answer $\Leftrightarrow$ null-grounding rule, query-disjoint splits, and redistributable image references. For InfoSeek, answer matching follows the official two-type protocol: string questions use relaxed exact match over the gold set; VALUE questions extract numbers from the prediction and accept any inside the gold range — a deterministic rule, not an LLM judge.

D Construction Pilot Statistics

A 100-query-per-benchmark construction pilot ($P=10$) produced 207 / 170 / 166 deduplicated trajectories and 464 / 366 / 355 step samples on InfoSeek / ScienceQA / GQA respectively, of which 426 / 328 / 341 survived QC (drops dominated by the grounding–outcome gap rule). The pilot surfaced — and motivated the fixes for — three construction-level failure modes: (i) *parametric lucky guesses* on InfoSeek (answers judged correct with zero evidence support), addressed by the support gate of Eq. (6) and illustrated in Appendix F, Case 2; (ii) *systematic entailment penalties* on ScienceQA reasoning questions (105

of the 170 answer steps flagged correct-but-unsupported), addressed by the NOT_REQUIRED verdict (Case 3); and (iii) *action imbalance and trajectory collapse* on InfoSeek (of its 426 retained samples, 206 are answer steps and the 220 retained search steps split 166:54 image:text — inverted against a text-only corpus; and only ≈ 2 deduplicated trajectories survive per query), addressed by retrieval-chain and proposal-diversity constraints in the policy prompt (Appendix E). We report these numbers for transparency: dataset papers rarely publish their construction pathologies, yet they determine label quality.

E Prompts

This appendix reproduces, verbatim, the three prompts used by the system: the policy proposal prompt (inference-time expansion and dataset rollouts), the rationale-generation prompt (dataset Step 3), and the answer-support judge prompt (dataset Step 2). Placeholders in braces are filled at run time.

E.1 Policy Proposal Prompt (Expansion / Rollouts)

```
You are planning retrieval actions to answer a question about the given image.

Question: {question}
Evidence collected so far:
{evidence}

Propose up to {k} candidate next actions as JSON, one per line. Allowed:
{"type": "text_search", "query": "..."} # query the corpus with text
{"type": "image_search"} # query the corpus with the image
{"type": "answer", "text": "..."}

Rules:
- The question refers to the given image (e.g. "this bird", "this building").
  Do NOT invent or assume a specific entity identity (a concrete species,
  landmark, person, ...) that the image and evidence have not established.
  Guessing an identity and searching for it propagates errors through the
  whole trajectory.
- Using an entity name that ALREADY APPEARS IN THE EVIDENCE above is NOT
  guessing: once image_search (or an earlier text_search) has surfaced the
  entity, you SHOULD build text_search queries from that evidence-established
  name.
- Retrieval chain for entity questions: if the entity is not yet identified,
  FIRST use image_search to match the image against the corpus. Once the
  entity is identified but the ASKED ATTRIBUTE is still missing from the
  evidence, the next action MUST include a text_search combining the
  evidence-established entity name with the asked attribute (e.g.
  "<entity name> opening date") -- do NOT answer from an entity match alone.
- Make the candidates genuinely different from each other: text_search
  queries within one proposal must not be near-duplicates -- vary the angle
  (entity name + attribute, bare keywords, a rephrasing of the question), and
  never re-issue a query already executed on this path.
- Keep every search query faithful to the original question's intent; do not
  drift away from what is actually being asked.
- Only answer when the collected evidence actually supports the answer, OR
  when the question is answerable from the image and question alone by
  perception, logic, or everyday commonsense (no external facts needed).
Output ONLY the JSON lines.
```

E.2 Rationale-Generation Prompt (Dataset Step 3)

```
You are writing a training rationale for a process reward model that judges
retrieval actions.

Question: {question}
Evidence collected BEFORE this action:
{evidence_before}
Action taken: {action_type}({action_input})
Evidence retrieved BY this action:
{observation}
Assigned quality score: {score} (0 = poor action, 1 = excellent action)

Write a rationale of 30-150 tokens explaining WHY this score is appropriate.
Requirements:
- explicitly mention the action type "{action_type}";
- cite at least one evidence id in square brackets, e.g. [e0];
- be concrete about how the evidence supports (or fails to support) progress
  toward answering the question;
```

```

- judge the step ONLY from the question, the evidence, and the action -- as if
the final outcome were still unknown. NEVER mention or hint at a gold /
ground-truth answer and never use evaluation language such as "the answer is
correct" or "matches the ground truth";
- if the evidence does not actually support the step's usefulness, say so
plainly.
After the rationale, end with ONE final line of exactly this form (it must be
consistent with the assigned score):
VERDICT: good      (clearly useful step)    or
VERDICT: mixed     (partially useful)      or
VERDICT: poor      (useless or harmful)

```

E.3 Answer-Support Judge Prompt (Dataset Step 2)

```

Question: {question}

Evidence collected so far:
{evidence}

Proposed final answer: {answer}

Classify how this answer relates to the evidence. Reply with exactly one word:

- SUPPORTED: the evidence explicitly contains or entails the answer. For a
multiple-choice answer it is enough that the evidence entails the CONTENT of
the chosen option (verbatim wording is not required). An absence answer like
"no" counts as supported when the evidence describes the relevant
scene/content and it lacks the asked-about item.
- PARTIAL: the evidence points toward the answer but leaves a substantive part
of it unconfirmed.
- UNSUPPORTED: the answer relies on external factual knowledge (entity facts,
numbers, dates, names, statistics) that the evidence does not contain.
- NOT_REQUIRED: the answer is derivable from the question text and the visible
image content alone, by perception, logic, or everyday commonsense -- no
external factual knowledge is needed. Do NOT use this for fine-grained
entity facts (species data, dates, measurements, biographies): those always
require evidence.

```

F Case Studies from the Construction Pilot

The following four cases are drawn from real trajectories of the construction pilot (Appendix D); evidence snippets are abridged. They illustrate, in order: the intended behavior, and the three failure modes that shaped the reward design of Section 3.3.

Case 1 (InfoSeek, success): the disambiguate-then-look-up chain

- **Question:** “When did this building officially open?” (image of the Pro Football Hall of Fame); gold: September 7, 1963.
- **Step 1** TEXT_SEARCH(“opening date Pro Football Hall of Fame”) → passage containing “...opened on September 7, 1963...”. High grounding; on a successful trajectory; kept as a high-score positive.
- **Step 2** IMAGE_SEARCH(query image) → further pages on the same entity (award lists, inductees). Topically related but adds no new information for the asked attribute — see Case 4.
- **Step 3** ANSWER(“September 7, 1963”) → correct *and* supported: the date is present in cited evidence, so $\text{supp}=1$ and the answer step receives a full-score label.

Case 2 (InfoSeek, failure): the parametric lucky guess

- **Question:** “What is the maximum weight this bird usually grows to?” (image of a mourning dove); gold: ≈ 153 g with tolerance.
- **Step 1** TEXT_SEARCH(“how heavy does mourning dove grow up to”) → passages comparing dove species qualitatively; no weight figure. **Step 2** IMAGE_SEARCH → sparrow and dove-call passages; irrelevant. Both search steps have very low grounding on a trajectory whose outcome is 1.0; the $|g - R_{\text{out}}| > 0.7$ rule drops both samples as self-contradictory labels.
- **Step 3** ANSWER(“Up to 145 grams”) → judged *correct* by the tolerance metric, yet *no evidence contains any weight figure*. Without the support gate this step would be labeled 1.0 — a perfect score for answering from parametric memory — and the PRM would learn that evidence-free answering is optimal. With Eq. (6): $\text{supp}=0$ (the unmatched number vetoes), the label

drops to $\gamma \cdot 1.0 = 0.3$, and the sample is flagged `unsupported_correct` for use as a DPO rejected side against sibling actions that kept searching.

Case 3 (ScienceQA, failure): entailment-gating a reasoning answer

- **Question:** “Which question can Jen’s experiment best answer?” (options about radish size vs. leaf count under different light); gold: the size option.
- **Step 1** TEXT_SEARCH(“radishes bright light dim light growth”) → a passage describing the experiment comparing radish *sizes* under two light conditions — decisive for choosing the size option.
- **Step 2** ANSWER(size option) → correct. A pure entailment judge, however, rules UNSUPPORTED: no document states “therefore choose option A” — the answer is *derived* from the evidence and the option semantics, not contained in any span. In the pilot, 105 of the 170 ScienceQA answer steps were flagged correct-but-unsupported this way, pinning them at the 0.3 floor and teaching the PRM a systematic bias against reasoning-type answers. The NOT_REQUIRED verdict (Appendix B) repairs exactly this: the judge classifies the question as answerable from the question, options, and evidence-backed premises by reasoning, and the answer step falls back to its outcome label.

Case 4 (open issue): redundant retrieval on solved states

- In Case 1, Step 2 retrieves same-entity content *after* the asked attribute is already in the bundle. Its grounding is not low — same-entity results are topically relevant — and it lies on a successful trajectory, so it receives a high label from Eq. (5) despite adding nothing: relevance and *marginal utility* diverge. This over-search pattern (cf. Wu et al. 38) is the one construction pathology we deliberately did not patch yet: a naive novelty discount $g' = g \cdot (1 - \max \cos(\text{new evidence}, \mathcal{E}))$ would also punish legitimate multi-source corroboration, which the poisoning robustness of Section 5.3 depends on. We record a marginal-gain field per step in ETBench-Open and defer the labeling decision until the full-scale distributions separate redundancy from corroboration; same-state DPO pairs (answer-now vs. search-more, both successful) provide a label-free training route for *when to stop* in the meantime.

G Notation

Symbol	Meaning
q, I	question; optional query image
$s_t, \mathcal{E}_t$	state; evidence bundle (Eq. 1)
$\mathcal{A}$	typed action space (Eq. 2)
$\tau, y(\tau), y^*$	trajectory; its answer; gold answer
$g(a, s)$	local grounding (Eq. 3)
$R_{\text{out}}(s, a)$	tree-level outcome credit (Eq. 4)
$\text{supp}(y, \mathcal{E})$	answer-support verdict (Eq. 6)
α, γ	fusion weight (0.5); support floor (0.3)
$Q_\phi(a s)$	PRM prior value for a candidate action
$Q(a s)$	running mean of backed-up rewards (Eq. 7), seeded by Q_ϕ
$N(s), N(s, a)$	node / edge visit counts
c	UCB1 exploration constant (1.0)
P, d_{max}, k	rollouts (10); max depth (3); children (3)
η_{stop}	early-stop threshold

Table 5: Notation.